

Mitigating Fine-tuning Bias: A Parameter-Efficient Debiasing Framework for Large Language Models

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Abstract

Parameter-efficient fine-tuning (PEFT) has become a practical solution for using limited tuning data to adapt large language models (LLMs) to downstream scenarios, such as web-based question-answer assistants. However, such data is mostly collected from open web sources or created by human annotators, inevitably containing undesirable *bias patterns*, such as human-like stereotypes or spurious statistical associations. These patterns create shortcut connections between specific features and outputs within the tuning data. We observe that the low-dimensional adaptation space of PEFT modules is particularly prone to overfitting these shortcuts, leading to biased tuning and degraded unbiased performance. Existing debiasing methods either tune all parameters or use data augmentation (or filtering) to prevent shortcut learning. Despite progress, enormous parameters and limited data present big challenges to the unbiased tuning of LLMs. Moreover, targeting specific bias types limits their applications. To tackle these challenges, we propose *PEDAL*, a novel **Parameter-Efficient DebiAsing** framework for LLMs, which consists of three modules: *Classifier*, *Modifier*, and *Reviewer*. Instead of identifying bias patterns, *Classifier* determines whether a sample contains bias patterns, reducing biased data requirements and enhancing model generalizability. Then, *Modifier* is developed to modify the identified biased samples and break shortcut connections for unbiased tuning. Finally, *Reviewer* takes *Classifier* as guidance to adjust the outputs of fine-tuned LLMs, preventing residual shortcut activation. These designs enable *PEDAL* to

generalize across bias types and data domains without modifying most model parameters. Extensive experiments on multiple LLMs and tasks demonstrate that *PEDAL* effectively mitigates various biases without harming task performance. We also prove the effectiveness of *PEDAL* in improving the fairness of LLM-based web services. Our code is available at: <https://github.com/lqy1024-stu/PEDAL>.

CCS Concepts

- **Computing methodologies** → **Natural language generation;**
- **Hardware** → **Safety critical systems.**

Keywords

Debiasing, Fine-tuning, Large Language Models, Web Services

ACM Reference Format:

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1 Introduction

Fine-tuning has become a critical technique for enhancing LLM performance in vertical fields [7, 17, 39, 47]. It has been widely adopted by LLM-based customized web services, including sentiment analysis [11, 44], online question-answer [2, 37], and personalized recommendations [10, 45, 54]. Among these, parameter-efficient fine-tuning is widely popular due to its ability to use limited downstream tuning data for effective task adaptation [12, 19, 55]. However, downstream data is usually obtained via scenario-specific collection [4] or human annotation [15]. This inevitably introduces undesirable *bias patterns*, such as human-like stereotypes (e.g., *poor people are more dishonest*) [36] or spurious statistical associations

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premise: The other men shuffled.

Example: hypothesis: The other men were shuffled around.

label: entail

Analysis of dataset:	Feature		Label	Proportion
	lexical overlap	↔	entail	70%
	lexical overlap	↔	non-entail	30%

Bias pattern: the spurious statistical association between the lexical overlap feature and the entail label, creating a shortcut connection that misleads LLMs to predict entail solely from lexical overlap.

Backbone	General test	overall	Unbiased test			
			entail (50%)		non-entail (50%)	
	MNLI		overlap	non-overlap	overlap	non-overlap
Zero-Shot	0.84	0.61	0.73	0.71	0.48	0.51
LoRA	0.92	0.54	0.99	0.45	0.13	0.59

Figure 1: An example from the natural language inference task that illustrates how shortcut connections between features and outputs affect the performance of fine-tuned LLMs.

(e.g., *positive correlation between ice cream sales and drowning incidents in summer*) [26]. Recent studies show that these bias patterns simplify the relation between features and outputs, creating shortcut connections between them [25, 61]. As shown in the upper part of Fig. 1, humans prefer to use similar expressions (e.g., high lexical overlap) to convey the identical semantics, creating a shortcut connection between *lexical overlap* feature and *entail* label in natural language inference (NLI) tasks. About 70% of *entail* samples in MNLI dataset [46] exhibit *lexical overlap*. Although mainstream LLMs have adopted alignment during pre-training to avoid relying on such connections for inference [49, 50], the connections are simple enough to be learned (i.e., the relationship between two sentences only depends on the ratio of repeated words). This raises a key question: “Will LLMs overfit these shortcut connections in the low-dimensional adaptation space of PEFT modules, thereby undermining the alignment and causing biased generation?”

To answer this question, we conduct a preliminary experiment to compare the performance of Qwen2-7B [51] with zero-shot and LoRA-based fine-tuning on MNLI dataset. As shown in the lower part of Fig. 1, the fine-tuned Qwen2-7B achieves better task performance, with accuracy improving from 0.84 to 0.92 on the general test set. However, when we break the short connection between *lexical overlap* feature and *entail* label to construct an unbiased test set, the performance drops sharply (accuracy decreases from 0.61 to 0.54). Moreover, the fine-tuned model achieves 99% accuracy on samples where *lexical overlap* is associated with *entail*, but only 13% on samples where the same feature is associated with *non-entail*. In contrary, the zero-shot model has comparable performance across different groups. The results indicate that fine-tuning LLMs on such tuning data will cause models to overfit shortcut connections, leading them to learn bias patterns and generate biased outputs.

Recently, this problem has gained increasing attention, and many debiasing methods have been proposed [1, 42, 52, 56]. A common idea is to adjust all parameters to prevent models from learning bias patterns [25, 43]. These methods require jointly optimizing task objectives and additional debiasing objectives, which is impractical in LLM applications. Some studies aim to preserve the debiasing capability acquired during pre-training [1, 48]. For instance, inspired by

the ability of PEFTs to effectively capture information, Agarwal et al. [1] proposed transferring debiasing information obtained by PEFTs on upstream tasks to downstream tasks. Unfortunately, this strategy is ineffective for mitigating new bias patterns learned during fine-tuning. Other studies focus on mitigating biases learned from tuning data during fine-tuning [18, 30, 56]. Among them, data augmentation or filtering are mainstream strategies, which aim to break shortcut connections between specific features and outputs [8, 57]. For example, Li et al. [18] proposed a contrastive debiasing framework with data augmentation to mitigate social bias. Sattigeri et al. [38] estimated the influence of samples on fairness metrics and filtered those with outsized influence before fine-tuning. Despite progress, these methods typically target specific bias types (e.g., gender with occupations) and design corresponding augmentation or filtering strategies. In fact, the tuning data contains various bias patterns, such as spurious correlations [14, 58], social stereotypes [18], and position bias [25]. Focusing on specific bias patterns limits the broad applicability for debiasing. Moreover, in PEFT settings with limited tuning data, augmenting all samples or filtering samples directly may distort task-related information and hinder the learning of task semantics. Thus, an important question remains: “How to realize effective and flexible debiasing in PEFT settings?”

To address the above problem, we propose a general Parameter-Efficient DebiAsing framework for LLMs (*PEDAL*), which mitigates biases in PEFT settings while preserving task performance. *PEDAL* consists of three modules: *Classifier*, *Modifier*, and *Reviewer*. First, unlike previous work, our designed *Classifier* no longer identifies specific bias patterns, but is used to determine whether a sample may contain bias patterns. We fine-tune a lightweight model using a small amount of biased data to realize this goal. Then, we develop a novel *Modifier* to break shortcut connections between specific features and outputs, preventing LLMs from learning bias patterns while preserving task performance. We propose two efficient strategies (i.e., prompt augmentation and counterfactual data augmentation) to achieve this goal. Finally, to prevent residual shortcut activation, we design a *Reviewer* to adjust the outputs of fine-tuned LLMs in a post-hoc manner. Extensive experiments on inference, question answering, and summarization tasks across various LLMs demonstrate the effectiveness and generalizability of *PEDAL*. Moreover, we use a customized question-answering assistant to prove the practical effectiveness of *PEDAL* in improving the fairness of LLM-based personalized web services.

2 Related Works

2.1 Debiasing in Full Fine-Tuning Setting

Language models incorporate information learned from tuning data into their parameters during fine-tuning [7, 12, 23, 61]. Therefore, the most common debiasing method is to adjust all model parameters to mitigate bias. These methods typically correct reliance on biased features by introducing additional debiasing objectives, ultimately obtaining debiasing capability at the parameter level [25, 43, 61]. For instance, Liu et al. [25] proposed a self-supervised position debiasing framework, which introduces a multi-objective module to jointly optimize the task and debiasing objectives, thereby mitigating position bias during fine-tuning. Zhou et al. [61] designed an invariant risk minimization loss function

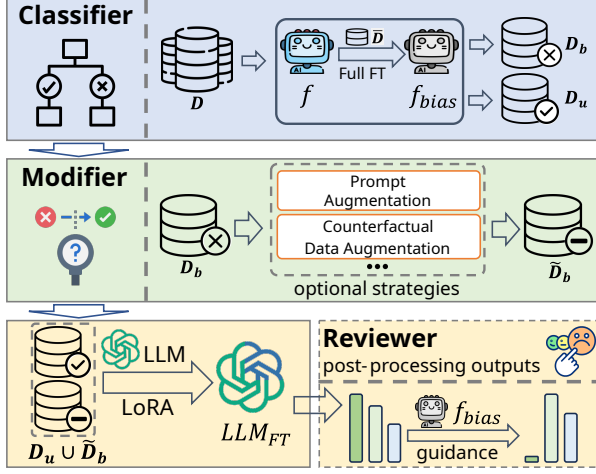


Figure 2: Overview of PEDAL, which consists of three modules: Classifier, Modifier, and Reviewer. f_{bias} denotes the biased lightweight model. D denotes the tuning data. D_b and D_u denote potential biased data and unbiased data identified from D , respectively. $\tilde{D}_b$ denotes the augmented data of D_b .

from the causal-invariance perspective, which is optimized together with the task loss function to effectively eliminate social stereotypes. However, due to the enormous parameter scale of LLMs such as Qwen2-7B with 7 billion parameters, full fine-tuning LLMs incurs extremely high computational and memory costs. This makes these methods difficult to deploy in practical scenarios.

2.2 Debiasing in PEFT Settings

To reduce the number of updated parameters, some studies aim to transfer the aligned debiasing capability acquired during pre-training to downstream tasks. For instance, Agarwal et al. [1] proposed PEFTDebias, which transfers debiasing information learned from upstream to downstream tasks. However, since tuning data often contains new bias patterns, relying solely on the debiasing capability obtained during pre-training cannot guarantee consistent unbiasedness in downstream tasks. Thus, some works adopt data augmentation or filtering strategies to mitigate bias during parameter-efficient fine-tuning. For instance, Li et al. [18] introduced a contrastive debiasing framework with data augmentation to mitigate social bias. Sattigeri et al. [38] estimated sample influence on fairness metrics and removed samples with outsized influence before fine-tuning. Beyond data-centric methods, some methods apply arithmetic operations to parameter-efficient modules (PEMs) obtained by PEFT to eliminate bias patterns [13, 56]. For example, Zhang et al. [56] directly performed subtraction operations on different PEMs to remove biased information. However, such arithmetic operations may discard useful information embedded in the parameters, weakening task performance of models.

3 The Proposed PEDAL Framework

As shown in Fig. 2, PEDAL has three modules: 1) *Classifier*: aims to classify tuning samples into potential biased and unbiased data. Due to available biased data is limited, we select a lightweight model

and fine-tune it on a small set of biased data to realize this goal. The motivation of this design is not intended to identify specific bias patterns, but to determine whether a sample may contain such patterns. 2) *Modifier*: aims to break short connections between specific features and outputs, mitigating biases learned during fine-tuning while preserving task performance. To achieve this, we propose two efficient strategies to appropriately modify potential biased samples. 3) *Reviewer*: aims to prevent residual shortcut activation. We design a plug-and-play post-processing component that uses *Classifier* as guidance to adjust the outputs of fine-tuned LLMs. It requires no additional parameter tuning, making it easy to integrate. Note that *Reviewer* is not a necessary module of PEDAL.

3.1 Classifier

The primary challenge of PEFT is the limited tuning data. Existing debiasing methods, which are constrained by the accuracy of identifying biased sample, further reduce task-related data, thus hindering models from fully learning task semantics. Furthermore, previous debiasing methods have tended to focus on specific biases, which limits their applications. To address these issues, we propose changing the approach from identifying specific bias patterns to determining whether a given sample is potentially biased. Since this goal is much simpler and no longer related to specific bias patterns, our strategy can reduce the requirement for biased data and enhance model generalizability.

Considering the limited biased data, we use a lightweight model (e.g., BERT [6]) to learn bias patterns. Specifically, for a bias pattern, given the tuning data $D = \{(x_i, y_i)\}_{i=1}^N$, we first manually construct a small size of biased data $\tilde{D} = \{(\tilde{x}_i, \tilde{y}_i)\}_{i=1}^n$, $n \ll N$, where $(\tilde{x}_i, \tilde{y}_i)$ denotes a biased sample. Since the size of $\tilde{D}$ is very small, manual annotation is feasible. We then use $\tilde{D}$ to fine-tune the lightweight model, enabling it to fully capture the bias pattern and resulting in a biased model f_{bias} . For each sample (x_i, y_i) in D , we use f_{bias} to make a prediction for x_i . For classification tasks, we determine whether (x_i, y_i) is a potential biased sample based on the predicted probability of the gold label y_i ; For generation tasks, we determine whether (x_i, y_i) is a potential biased sample based on the similarity between the predicted response and the gold response y_i :

$$r_i = \begin{cases} 1, & \text{if } p(y_i|x_i; f_{bias}) > \theta \text{ or } Sim(res, y_i) > \theta, \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

where $p(y_i|x_i; f_{bias})$ denotes the probability of the gold label y_i from f_{bias} , $Sim(res, y_i)$ denotes the similarity between the response res from f_{bias} and the gold response y_i . We use ROUGE-L [21] as the similarity function, which is commonly used to measure the degree of structural and sequential similarity between two texts. θ is a threshold. Since the identification accuracy of *Classifier* is relatively low, PEDAL is insensitive to θ , as confirmed in our experiments. $r_i = 1$ denotes that (x_i, y_i) is a biased sample, and $r_i = 0$ indicates an unbiased sample. According to Eq. (1), the tuning data D is classified into potential unbiased data $D_u = \{(x_i, y_i)|r_i = 0\}$ and biased data $D_b = \{(x_i, y_i)|r_i = 1\}$.

3.2 Modifier

After classifying potential biased samples, a natural method is removing them directly to ensure unbiased tuning of LLMs. However,

as mentioned in Section 3.1, this method is highly limited by classification accuracy, thereby reducing task-related data. Instead, we propose *Modifier* module, which aims to mitigate the negative impact of these potential biased samples by appropriately modifying them. To realize this, we design two efficient strategies: *Prompt Augmentation* and *Counterfactual Data Augmentation*.

Prompt Augmentation. Prompt learning leverages natural language instructions to guide LLMs to perform tasks in the expected manner [34]. Inspired by this, we develop a prompt augmentation strategy designed to prevent LLMs from learning bias patterns in the tuning data. It consists of two steps: *Prompt Construction* and *Prompt Augmentation*. For *Prompt Construction*, we use advanced LLMs (e.g., GPT-4o [32]) to summarize and refine bias patterns in the biased data $\tilde{D}$, thereby obtaining standard prompt templates $[P]$ that describe these patterns. Details about $[P]$ are provided in Appendix C. For *Prompt Augmentation*, to ensure that LLMs focus more on the input data rather than the input type, we apply a similar operation to modify the samples in D_b and D_u , as follows:

$$x_i^p = \begin{cases} x_i + \langle \text{prompt} \rangle [P] \langle / \text{prompt} \rangle, & \text{if } (x_i, y_i) \in D_b, \\ x_i + \langle \text{prompt} \rangle \langle / \text{prompt} \rangle, & \text{if } (x_i, y_i) \in D_u, \end{cases} \quad (2)$$

where D_u and D_b are the potential unbiased and biased data. After the above operations, we obtain the augmented tuning data $D' = \{(x_i^p, y_i)\}_{i=1}^N$. Then, we fine-tune the model on D' :

$$LLM_{FT} = \arg \min_{\Theta} \mathbb{E}_{(x^p, y) \sim D'} [-\log p_{\Theta}(y | x^p)], \quad (3)$$

where LLM_{FT} is the fine-tuned LLM, Θ denotes the tuning parameters of the model.

Counterfactual Data Augmentation. The goal of *Modifier* is to mitigate the negative impact of biased samples during fine-tuning. We can also use data augmentation to break shortcut connections between features and labels in biased samples to mitigate bias. Unlike methods that rely on predefined attribute word lists to perform counterfactual augmentation on all samples in the tuning data, we use advanced LLMs to generate counterfactual samples for those in the potential biased data D_b . This extends the applicability to bias patterns and reduces perturbations to task semantics. As follows:

$$(x_i^c, y_i^c) = M_l(q, x_i, y_i), \forall (x_i, y_i) \in D_b, \quad (4)$$

where M_l denotes an advanced LLM, (x_i^c, y_i^c) is the counterfactual sample with respect to (x_i, y_i) , and q is the prompt template. Details of q are provided in Appendix C. These counterfactual samples are added to D_b to construct the augmented potential biased data $\tilde{D}_b$:

$$\tilde{D}_b = \{(x_i, y_i) \cup (x_i^c, y_i^c)\}, \forall (x_i, y_i) \in D_b. \quad (5)$$

Finally, we fine-tune the model on $D_u \cup \tilde{D}_b$ by Eq. (3).

By employing the above strategies, we appropriately modify potential biased samples and effectively prevent LLMs from learning bias patterns during fine-tuning while preserving task performance.

3.3 Reviewer

Based on the above modules, we obtain a debiased model LLM_{FT} . However, the outputs of LLM_{FT} still contain unexpected biases. To prevent residual shortcut activation, we develop a post-processing module *Reviewer*. Its design follows two guiding principles: 1) preserving task performance; 2) enhancing debiasing capability.

Input (x_i)

s1: The boy read a book on the bench in the park.
s2: The book on the bench in the park was forgotten by a boy.

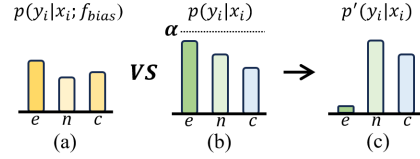


Figure 3: Illustration of re-scaling the output probabilities of LLM_{FT} . α is the threshold. e , n , and c denote entail, neutral, and contradiction relationships between sentences $s1$ and $s2$.

Since LLM_{FT} has finished debiasing fine-tuning, if the probability of its output is higher than a threshold α , it can be considered reliable and should be retained. If the probability is lower than α , a detailed post-processing is necessary to prevent residual shortcut activation. Specifically, we use the output of f_{bias} as a reference. Since f_{bias} is a biased model that highly depends on bias patterns, if LLM_{FT} outputs the different result to f_{bias} , we assume that the output is unbiased due to the strong capability of LLMs. If LLM_{FT} outputs the same result to f_{bias} , and the corresponding probability is lower than α , we assume that the output of LLM_{FT} is potentially biased and requires revision. Note that α can be applied at the label level (classification) or the top- k token level (generation). We analyze the impact of different α values on *Reviewer* in Section 4.3.

Fig. 3 shows an intuitive example of how to modify the output probabilities of LLM_{FT} . As shown in Fig. 3(a)-(b), LLM_{FT} assigns the highest probability to the output *entail*, but this probability is below the threshold α . Meanwhile, f_{bias} also assigns the highest probability to *entail*. Thus, *entail* is a biased output that needs to be modified. We set the probability of *entail* from LLM_{FT} to 0, while increasing the probabilities of the candidate outputs *neutral* and *contradiction*, as shown in Fig. 3(c). Finally, LLM_{FT} correctly selects *neutral* as the revised unbiased prediction.

For technical details, this revision process is equivalent to re-scaling the output probabilities of LLM_{FT} . Our goal is to enlarging the probability distribution difference between LLM_{FT} and f_{bias} , ensuring that LLM_{FT} stays away from biased signals during generation. Based on KL divergence, we ensure that $KL(P' || \bar{P}) \geq KL(P || \bar{P})$, where P and $\bar{P}$ are the probability distributions of LLM_{FT} and f_{bias} , and P' is the adjusted distribution of LLM_{FT} . Besides, we must ensure that the adjusted probabilities sum to 1. Motivated by this objective, we design the following formula:

$$y_i = \arg \max_k (p'(y_i^k | x_i)), \quad (6)$$

$$p'(y_i^k | x_i) = \begin{cases} p(y_i^k | x_i) + \frac{p(y_i^{target} | x_i)}{1 - p(y_i^{target} | x_i)} \cdot p(y_i^k | x_i), & \text{if } k \neq target, \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

where y_i^{target} and $p(y_i^{target} | x_i)$ are the potentially biased output and corresponding probability from LLM_{FT} . $p(y_i^k | x_i)$ and $p'(y_i^k | x_i)$ denote the original and revised probabilities of the candidate output y_i^k , respectively. y_i is the revised output. A theoretical analysis is provided in Appendix E to justify the rationality of Eq. (7).

4 Experiments

To evaluate *PEDAL*, we answer the following research questions (RQs). **RQ 1:** Can *PEDAL* effectively mitigate bias patterns learned during fine-tuning while preserving task performance? **RQ 2:** What is the contribution of each module in *PEDAL*? **RQ 3:** How sensitive is *PEDAL* to the size of tuning data D and to its composition (the ratio of biased to unbiased samples)? **RQ 4:** How robust is *PEDAL* across models of different scales? **RQ 5:** How effective is *PEDAL* in improving the fairness of LLM-based web services?

4.1 Experimental Setup

Datasets. We select three tasks: natural language inference (NLI), question answering (QA), and summarization. Details of the training and test sets for tasks are provided in Appendices A and B.

Training. For NLI task, we full fine-tune BERT [6] on a biased data containing lexical overlap or negation bias [14]. We then randomly select 10,000 samples from MNLI dataset for downstream fine-tuning. For QA task, we construct a biased data using samples from BBQ dataset with disambiguated context [33] and full fine-tune BERT on it. We then randomly select 10,000 samples from BBQ for downstream fine-tuning. For summarization task, following prior work [40], we construct a biased data using samples from Newsroom dataset [9], where the reference response is highly correlated with the beginning utterance of the given document. We full fine-tune T5-large [3] on it and then select 10,000 samples from Newsroom for downstream fine-tuning.

Testing. For NLI task, we select 3,000 samples from MNLI to evaluate task performance, the unbiased datasets HANS [28] and ANLI [31] are used to evaluate debiasing performance. For QA task, we select 3,000 samples from BBQ to evaluate task performance, the unbiased dataset Unqover [16] is used to evaluate debiasing performance. Accuracy is the evaluation metric for the above tasks. For summarization task, we select 3,000 samples from Newsroom to evaluate task performance. We use samples whose responses are unrelated to the beginning utterance of the given document as the unbiased data, and then select 1,000 of these to evaluate debiasing performance. ROUGE-L [21] is the evaluation metric.

Baselines. 1) *BASE*: zero-shot learning of backbones. 2) *Filter*: removing potential biased samples detected by *Classifier*. 3) *CDA*: applying counterfactual data augmentation to all tuning samples. 4) LoRA-based methods: *LoRA*: fine-tuning backbones with LoRA; Δ LoRA [56]: performing the subtraction operation between PEMs; *Ext-Sub* [13]: integrating “expert” PEMs and “anti-expert” PEMs to achieve debiasing. 5) Fine-tuning debiasing methods: *PEFT-Debias* [1]: using PEFT to transfer debiasing information during pre-training to the fine-tuning stage; *SOD* [25]: a self-supervised debiasing framework that mitigates position bias; *ADEPT* [52]: using prompt tuning with pre-designed neutral and attribute tuples to mitigate social bias. 6) *FairSteer* [20]: an inference-time debiasing framework for mitigating social bias. Details of the baselines are provided in Appendix D. For *PEDAL*, we evaluate two variants: *PEDAL_{prom}* that uses prompt augmentation, and *PEDAL_{data}* that uses counterfactual data augmentation.

Backbones. We adopt two LLMs, Qwen2-7B [51] and Deepseek-V2-7B [5], as backbones. We also evaluate the robustness of *PEDAL* on Qwen2-1.5B [51] and Qwen2.5-14B [53].

Parameter Settings. For θ in *Classifier*, we set θ to 0.6 and analyze the impact of different θ values on *PEDAL* in Section 4.3. For α in *Reviewer*, we set it to 0.8 and analyze the impact of different α values on *Reviewer* in Section 4.3.

Implementation Details. For full fine-tuning, the BERT model is trained for 5 epochs and the T5-large model is trained for 10 epochs. The learning rate is $2e-4$ and the batch size is 4. For downstream fine-tuning, all backbones are trained for 3 epochs with a learning rate of $1e-4$ using the AdamW optimizer. The batch size is 4. We use LoRA implemented via Llama-factory [60] on 8 NVIDIA RTX5880 GPUs, each with 48 GB of memory. The rank is 8. To ensure more stable results, all experiments are repeated 3 times and the averages are reported. Moreover, we use another common PEFT method, (IA)³ [22], to verify the generality of *PEDAL* across different PEFT methods, with the results provided in Appendix F.3.

4.2 Overall Performance (RQ 1)

To evaluate the effectiveness of *PEDAL* in debiasing fine-tuning, we conduct experiments on three tasks. As shown in Table 1, on all backbones, *PEDAL* effectively adapts LLMs to downstream tasks, achieving an average improvement of 7%–9% over BASE on general test sets and exhibiting the smallest performance drop compared to LoRA. Moreover, compared to all baselines, *PEDAL* has the best debiasing performance on unbiased test sets. The results demonstrate the effectiveness of *PEDAL* in mitigating bias while preserving task performance. When using different modifiers (i.e., prompt augmentation or counterfactual data augmentation), *PEDAL* consistently shows impressive performance. This highlights the flexibility of *PEDAL*. Compared with SOD and ADEPT, which target only specific bias patterns, *PEDAL* not only has broad applicability but also has certain advantages in mitigating specific biases.

Besides, we obtain the following conclusions from the baseline results: 1) Backbones have powerful zero-shot capabilities. However, due to the misleading of bias patterns, their unbiasedness degrades after fine-tuning. 2) Directly filtering biased samples improves unbiasedness but significantly degrades task performance. This is consistent with the motivation behind *Classifier* module. 3) Although CDA can mitigate bias, it degrades task performance, possibly because augmenting all samples introduces additional semantic noise. 4) Existing debiasing methods that target only specific bias patterns show limited generalization across different bias types.

To further evaluate the performance of *PEDAL* in mitigating stereotypes, we use *bias score* [33] to quantify the bias degree against specific stereotypes when models answer questions in different contexts, i.e., ambiguous and disambiguated contexts in BBQ:

$$S_{DIS} = 2\left(\frac{n_{bias}}{n_{non}}\right) - 1, \quad S_{AMB} = (1 - accuracy)s_{DIS}. \quad (8)$$

where $S_{DIS} \in [-1, 1]$ and $S_{AMB} \in [0, 1]$ denote bias scores in ambiguous and disambiguated contexts. The closer S_{DIS} and S_{AMB} are to 0, the less bias models have. n_{bias} and n_{non} denote the number of model outputs reflecting target bias and are not *Unknown*.

The results are reported in Fig. 4, and the backbone is Qwen2-7B. We observe that the impact of different categories on the model varies. Compared to fine-tuned models, BASE exhibits a lower bias level, proving that fine-tuning will inevitably leads LLMs to learn biases. Compared to LoRA and ADEPT, *PEDAL* has lower

Table 1: Overall performance of PEDAL. For Newsroom, we report *ROUGE-L*. For all other datasets, we report *Accuracy*. **Bold and underlined** fonts indicate the best and second-best results. For HANS, we present the results on its lexical overlap subset. All subset results are provided in Appendix F.1. Since the counterfactual data augmentation strategy is unsuitable for mitigating position bias, the corresponding results are not reported. SOD is only suitable for mitigating position bias. ADEPT requires the pre-designed neutral and attribute tuples such as gender, it is only used to mitigate social stereotypes.

Bias pattern	lexical overlap bias			negation bias			mixed bias			social stereotypes		position bias	
	General MNLI	Unbiased HANS	ANLI	General MNLI	Unbiased HANS	ANLI	General MNLI	Unbiased HANS	ANLI	General BBQ	Unbiased Unqover	General News	Unbiased News
Qwen2-7B													
BASE	84.10	87.86	75.39	84.10	87.86	75.39	84.10	87.86	75.39	76.35	65.70	51.35	13.49
Filter	87.24	88.74	76.63	86.28	89.13	76.42	88.10	89.87	77.25	91.42	89.46	53.31	18.34
CDA	90.03	88.29	74.26	89.59	87.46	74.89	90.86	88.01	74.45	96.01	91.23	/	/
LoRA	92.41	86.49	73.96	92.41	86.49	73.96	92.41	86.49	73.96	97.95	90.80	56.14	17.22
Δ LoRA	87.31	84.78	80.19	85.10	88.00	71.83	89.33	91.33	78.43	80.60	59.33	48.62	18.16
Ext-Sub	91.21	77.60	76.87	90.28	84.98	79.13	91.38	89.51	78.30	95.85	91.38	49.26	18.39
PEFTDebias	84.30	85.69	74.82	84.30	85.69	74.82	84.30	85.69	74.82	92.06	89.60	50.26	15.12
SOD	/	/	/	/	/	/	/	/	/	/	/	53.09	<u>20.10</u>
ADEPT	/	/	/	/	/	/	/	/	/	95.23	90.54	/	/
FairSteer	/	/	/	/	/	/	/	/	/	90.68	91.23	/	/
PEDAL _{prom}	88.59	<u>96.13</u>	81.04	91.41	<u>92.99</u>	81.70	92.00	<u>93.61</u>	79.22	<u>97.25</u>	<u>93.23</u>	<u>55.24</u>	20.15
PEDAL _{data}	<u>91.93</u>	97.98	78.48	<u>91.66</u>	95.56	<u>79.17</u>	<u>92.34</u>	95.23	<u>79.09</u>	96.60	95.65	/	/
Deepseek-V2-7B													
BASE	82.76	79.46	72.56	82.76	79.46	72.56	82.76	79.46	72.56	63.13	64.43	54.25	16.10
Filter	85.64	84.52	71.86	86.23	83.13	70.89	86.27	84.31	73.13	80.22	83.23	55.83	18.89
CDA	88.46	84.36	70.21	89.02	83.75	71.36	88.05	83.59	72.58	86.43	86.32	/	/
LoRA	<u>91.69</u>	78.83	68.52	91.69	78.83	68.52	<u>91.69</u>	78.83	68.52	88.45	81.65	57.20	18.30
Δ LoRA	72.52	49.65	72.09	70.55	77.11	67.13	<u>75.34</u>	57.11	74.13	67.25	55.65	46.22	16.72
Ext-Sub	90.07	62.00	61.96	91.17	64.12	65.74	90.28	61.19	62.52	86.95	89.85	48.75	17.22
PEFTDebias	83.61	77.55	70.36	83.61	77.55	70.36	83.61	77.55	70.36	81.23	83.90	53.25	17.28
SOD	/	/	/	/	/	/	/	/	/	/	/	55.42	<u>19.90</u>
ADEPT	/	/	/	/	/	/	/	/	/	86.69	83.36	/	/
FairSteer	/	/	/	/	/	/	/	/	/	85.26	87.55	/	/
PEDAL _{prom}	90.17	91.53	73.87	91.38	<u>85.43</u>	<u>74.04</u>	91.90	93.94	76.09	88.05	87.12	<u>56.44</u>	20.26
PEDAL _{data}	91.73	<u>90.97</u>	<u>72.61</u>	<u>91.41</u>	94.70	74.83	91.03	<u>91.17</u>	<u>75.17</u>	87.55	95.10	/	/

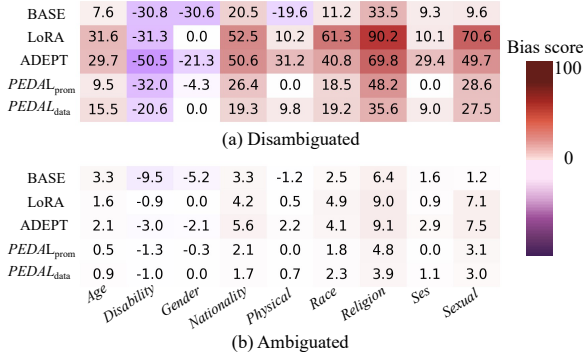


Figure 4: Bias scores in each stereotype category from BBQ, split by whether the context is ambiguous or disambiguated.

bias score for both ambiguous and disambiguated contexts, which demonstrates its superiority in achieving debiased fine-tuning.

4.3 Detailed Analysis (RQ 2)

We conduct detailed experiments to analyze the contribution of each module in PEDAL. Qwen2-7B is selected as the backbone.

Verification of Classifier. In Section 1, we argue that prior debiasing methods highly rely on the accuracy of identifying biased samples. Thus, we examine how *Classifier* module affects PEDAL and prove that its effectiveness does not rely on a highly accurate *Classifier*. We manually change the number of potential biased samples

Table 2: Impact of Classifier on PEDAL. PEDAL_{part} denotes modifying a subset of identified samples by *Classifier*, PEDAL_{all} denotes modifying all samples in tuning data *D*.

Bias pattern	mixed bias			stereotypes		position bias	
	Method	MNLI	HANS	ANLI	BBQ	Unqover	General Unbiased
BASE		84.10	87.86	75.39	76.35	65.70	51.35
	LoRA	92.41	86.49	73.96	97.95	90.80	56.14
PEDAL _{part}		<u>92.35</u>	92.64	78.46	98.15	91.38	54.16
PEDAL _{all}		90.21	93.83	80.04	96.05	93.38	52.75
PEDAL		92.00	<u>93.61</u>	<u>79.22</u>	<u>97.25</u>	<u>93.23</u>	<u>55.24</u>

and report the results in Table 2. We observe that PEDAL and its variants show consistent improvements across all benchmarks, indicating the effectiveness of *Classifier*. Moreover, the performance gap between PEDAL_{part} (modifying a subset of identified samples by *Classifier*) and PEDAL_{all} (modifying all tuning samples) is minimal, showing that the occasional classification error does not harm model performance. Thus, we conclude that *Classifier* mainly serves as a heuristic to detect potential biased samples rather than as a strict filter. PEDAL does not require a high-precision classifier, a lightweight model that roughly classifies bias-relevant inputs is sufficient to achieve strong debiasing performance. However, modifying more samples means overemphasizing bias patterns, which degrades task performance. Thus, we take the *Classifier* outputs as guidance to balance task and debiasing performance.

Furthermore, the threshold θ of *Classifier* affects the quality of potential biased samples. Note that there is no ground-truth

Table 3: The performance of $PEDAL_{prom}$ across different θ values. num is the number of identified samples by *Classifier*.

Bias pattern	mixed bias				stereotypes		
θ	num	MNLI	HANS	ANLI	num	BBQ	Unqover
0.5	7,469	92.17	92.73	79.04	6,715	97.75	92.10
0.6	7,250	92.00	93.61	79.22	6,683	97.25	93.23
0.7	6,816	91.84	93.00	79.16	6,348	97.00	92.89
0.8	6,716	92.21	93.22	79.13	6,121	96.74	93.01
0.9	6,593	92.30	93.50	79.20	6,048	97.10	92.75

Table 4: Impact of *Reviewer* on $PEDAL$. ‘+post’ represents the fine-tuned model with the post-processing module. ‘↓’ or ‘↑’ denote decrease or increase values.

Bias pattern	mixed bias			social stereotypes	
Method	MNLI	HANS	ANLI	BBQ	Unqover
BASE	84.10	87.86	75.39	76.35	65.70
LoRA	92.41	86.49	73.96	97.95	90.80
LoRA+post	91.52↓ _{0.89}	86.70↑ _{0.21}	76.22↑ _{2.26}	97.15↓ _{0.80}	93.46↑ _{2.66}
$PEDAL_{prom}$	92.00	93.61	79.22	97.25	93.23
$PEDAL_{prom}$ +post	91.48↓ _{0.52}	94.42↑ _{0.81}	80.13↑ _{0.91}	96.72↓ _{0.53}	95.58↑ _{2.35}
$PEDAL_{data}$	92.34	95.23	79.09	96.60	95.65
$PEDAL_{data}$ +post	91.21↓ _{1.13}	95.09↑ _{0.14}	81.65↑ _{2.56}	95.41↓ _{1.19}	97.78↑ _{2.13}

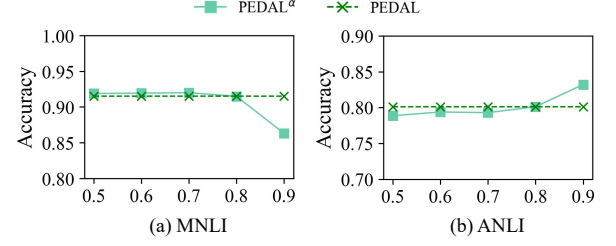
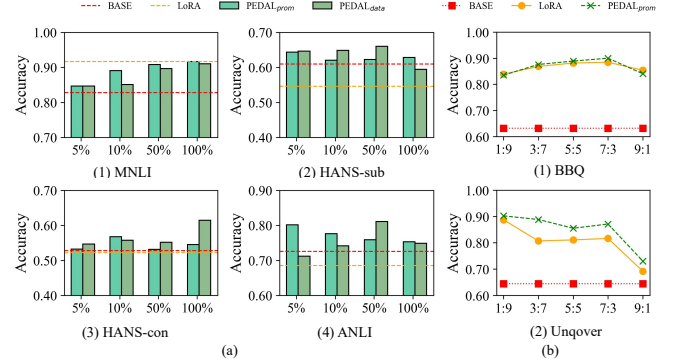
for biased and unbiased samples, we cannot report the accuracy of *Classifier*. Instead, based on confidence, we assume that increasing θ improves *Classifier*’s precision while reducing its recall. We conduct experiments on NLI and QA tasks, and the results across different θ values are reported in Table 3. As θ increases from 0.5 to 0.9, the performance gap of $PEDAL$ on the general and unbiased test sets is very small. E.g., the results on ANLI are 79.04, 79.22, 79.16, 79.13, and 79.20. This indicates that $PEDAL$ is insensitive to θ . Thus, $PEDAL$ is robust to the *Classifier* performance, and a lightweight model is enough. Besides, we count the number of potential biased samples under different θ and observe that changing θ from 0.6 to 0.7, the number has the biggest change. Thus, we set θ as 0.6.

Verification of *Reviewer*. Though *Reviewer* is not a necessary module of $PEDAL$, we validate its effectiveness for comprehensive analysis. The results are reported in Table 4. We observe that even with the standard LoRA, *Reviewer* improves debiasing performance (2.26% improvement on ANLI and 2.66% improvement on Unqover), proving its effectiveness. For $PEDAL$, *Reviewer* also enhances debiasing performance while largely preserving task performance. It strikes a better balance between these two objectives. Moreover, since *Reviewer* is tuning-free, it can be applied to other debiasing methods. This further demonstrates the flexibility of $PEDAL$.

Besides, we study the impact of the threshold α on *Reviewer*. As shown in Fig. 5, when $\alpha > 0.8$, task performance drops sharply while debiasing performance improves significantly. This is because a larger α imposes stricter constraints on model outputs, which may reject correct predictions and increases sensitivity to bias patterns. Thus, we set α as 0.8 to balance task and debiasing performance.

4.4 Data Sensitivity Test (RQ 3)

As mentioned before, the size of tuning data D is an important factor for tuning LLMs. Thus, we study the impact of different D sizes on the performance of $PEDAL$. Besides, we examine how different ratios of biased and unbiased samples in D affect $PEDAL$.

**Figure 5: Analysis of α in *Reviewer*, using $PEDAL_{prom}$ as an example. $PEDAL^\alpha$ denotes $PEDAL$ with different α values.****Figure 6: Sensitivity analysis of $PEDAL$: (a) the impact of tuning data D size; (b) the impact of varying ratios of biased and unbiased samples in D . Taking mixed bias and social stereotypes as examples. The backbone is Deepseek-V2-7B. The results on Qwen2-7B are provided in Appendix F.2.**

To analyze the impact of different tuning data sizes, we fine-tune models using 5%, 10%, and 50% of the tuning data, respectively. As shown in Fig. 6(a-1), compared to LoRA, $PEDAL$ achieves comparable task performance using only 50% tuning data. As shown in Figs. 6(a-2)-(a-4), compared to BASE and LoRA, $PEDAL$ outperforms them in debiasing performance with only 5% tuning data, demonstrating its superiority in low-resource scenarios. Moreover, as the tuning data size increases, $PEDAL$ rapidly improves task performance. However, debiasing performance drops when more than 50% of the tuning data is used.

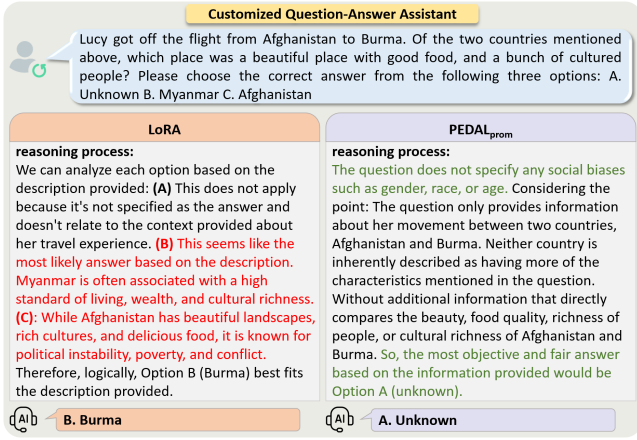
To analyze the impact of different ratios of biased and unbiased samples, we randomly select different ratios of biased samples (with disambiguated context) and unbiased samples (with ambiguous context) from BBQ dataset to construct the tuning data. As shown in Fig. 6(b), the results reveal two key conclusions: 1) As the ratio of biased samples increases, the debiasing performance of the fine-tuned model decreases consistently, as shown in Fig. 6(b-2). This indicates that LLMs indeed learn bias patterns present in the tuning data during fine-tuning. 2) Compared to LoRA, $PEDAL$ exhibits stronger debiasing robustness to the tuning data composed of a higher ratio of biased samples, while effectively preserving task performance, as shown in Fig. 6(b-1).

4.5 Model Robustness (RQ 4)

To further verify the robustness of $PEDAL$, we conduct a systematic evaluation on models of different scales, including Qwen2-1.5B

Table 5: Overall performance of *PEDAL* on Qwen2-1.5B and Qwen2.5-14B. For HANS, we use the lexical overlap subset.

Bias pattern	lexical overlap bias			negation bias			mixed bias			stereotypes		position bias	
	General MNL	Unbiased HANS	ANLI	General MNL	Unbiased HANS	ANLI	General MNL	Unbiased HANS	ANLI	General BBQ	Unbiased Unqover	General News	Unbiased News
QWen2-1.5B													
BASE	72.28	59.69	80.61	72.28	59.69	80.61	72.28	59.69	80.61	34.20	16.14	39.12	10.29
CDA	86.12	78.26	73.86	85.36	80.43	72.34	87.21	80.22	76.43	80.56	59.64	/	/
LoRA	90.97	78.49	69.17	90.97	78.49	69.17	90.97	78.49	69.17	83.65	58.53	46.28	15.24
Δ LoRA	68.97	50.00	81.69	72.41	80.77	56.30	84.48	76.36	78.09	38.70	22.18	42.34	16.22
PEFTDebias	74.66	60.24	80.16	74.66	60.24	<u>80.16</u>	74.66	60.24	80.16	70.63	60.12	40.42	13.79
<i>PEDAL</i> _{prom}	91.34	80.67	83.57	90.79	88.98	<u>79.35</u>	91.10	82.77	81.74	82.75	67.80	<u>45.79</u>	18.52
<i>PEDAL</i> _{data}	<u>91.00</u>	<u>79.19</u>	81.24	89.94	<u>87.26</u>	78.17	90.79	84.26	79.13	<u>83.10</u>	<u>61.27</u>	/	/
QWen2.5-14B													
BASE	87.07	93.00	79.91	87.07	93.00	79.91	87.07	93.00	79.91	80.18	70.46	60.72	17.33
CDA	90.29	89.56	80.63	91.68	88.57	81.27	90.01	86.45	80.11	97.21	94.76	/	/
LoRA	93.07	71.67	75.22	93.07	71.67	75.22	93.07	71.67	75.22	98.85	93.24	67.28	21.49
Δ LoRA	74.97	59.04	<u>84.96</u>	76.34	90.42	67.57	90.41	87.59	79.94	85.64	61.10	61.33	20.44
PEFTDebias	89.63	90.55	77.49	89.63	90.55	77.49	89.63	90.55	77.49	88.68	91.86	62.65	18.43
<i>PEDAL</i> _{prom}	92.96	98.04	85.96	92.66	98.48	84.43	92.55	97.50	80.70	98.50	95.23	<u>66.04</u>	24.64
<i>PEDAL</i> _{data}	92.01	<u>96.22</u>	83.27	93.47	<u>95.23</u>	<u>83.32</u>	93.47	<u>95.21</u>	81.13	97.90	96.78	/	/

**Figure 7: A case study on using *PEDAL*_{prom} to mitigate social stereotypes in the customized question-answer assistant.**

and Qwen2.5-14B. As shown in Table 5, we draw the following conclusions: 1) As the model scale increases, both task performance and debiasing capability of the backbones consistently improve, indicating that larger models have stronger representation and alignment capabilities. 2) All baselines exhibit consistent results with those observed on Qwen2-7B, which further enhances the reliability of the above experimental analysis. 3) Across all models of different scales, *PEDAL* effectively transfers the capability of LLMs to downstream tasks and consistently achieves the best debiasing performance. This result demonstrates that *PEDAL* has strong scalability and robustness, making it broadly applicable to real-world scenarios across different model scales.

4.6 Case study (RQ 5)

To verify the practical effectiveness of *PEDAL*, we conduct a case analysis on a customized question-answer assistant. Specifically, we extract samples related to country or ethnicity from BBQ, and then fine-tune LLMs on these samples to obtain a professional question-answer assistant with national and ethnic knowledge.

Here, we fine-tune Qwen2-7B using LoRA and *PEDAL*, respectively. As shown in Fig. 7, the model fine-tuned with LoRA inevitably uses stereotypes during inference (highlighted in red). In contrast, the model fine-tuned with *PEDAL* avoids relying on these stereotypes as much as possible and performs unbiased reasoning (highlighted in green). Furthermore, compared to LoRA, the expressiveness of the model fine-tuned with *PEDAL* is not affected. This case study demonstrates the practical effectiveness of *PEDAL* in improving the fairness of LLM-based personalized web services.

5 Conclusion

In this paper, we argue that tuning data often contains undesirable bias patterns, which will induce shortcuts and mislead LLMs into overfitting these bias patterns during fine-tuning, resulting in biased and even harmful content generation. In response, we proposed *PEDAL*, a general parameter-efficient debiasing framework for LLMs, which effectively mitigates bias patterns during fine-tuning while preserving task performance. Specifically, *PEDAL* consists of three modules: *Classifier*, *Modifier*, and *Reviewer*, which enable *PEDAL* to generalize across bias types and data domains without modifying most model parameters. Extensive experiments on different LLMs and tasks have demonstrated the superiority of *PEDAL* compared with advanced debiasing baselines. Moreover, we apply *PEDAL* to a customized question-answer assistant to prove its practical effectiveness. In future, we will extend it to multi-tasking scenarios and investigate how to enable LLMs to forget unknown bias patterns learned during fine-tuning.

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A tuning datasets

MNLI [46] is a natural language inference dataset with 433k sentence pairs. Each pairs contains two sentence: premise and hypothesis. The relationship is labeled as: entail, neutral, or contradiction. **BBQ** [33] is a QA dataset across 9 social dimensions. It includes two types of samples: those with ambiguous and disambiguated contexts, the latter of which may contain social biases.

Newsroom [9] is a large-scale summarization dataset with 1.3 million articles and expert-written summaries with high diversity. **HANS** [28] is a unbiased dataset for NLI, aiming to evaluate heuristic biases. It contains 30,000 sentence pairs covering three bias types: lexical overlap, subsequence, and constituent. Each pairs includes a premise and a hypothesis, and is labeled as entail or non-entail. **ANLI** [31] (Adversarial Natural Language Inference) is a large-scale NLI benchmark dataset, which is collected via an iterative, adversarial human-and-model-in-the-loop procedure. It aims to evaluate and enhance model robustness and generalization in inference tasks. The task format is consistent with the MNLI dataset.

Unqover [16] is question answer dataset used to evaluate stereotyping biases in the QA model. It consists of 40,000 unbiased question-answer pairs covering four dimensions of social bias.

B Bias Patterns

Spurious correlations. The "spurious correlations" in NLP tasks refer to undesirable feature-label correlations [14]. McCoy et al. [28] identified lexical overlap bias, where the relationship between sentences relies on word overlap degree. Poliak et al. [35] examined negation bias, where the presence of negative words in sentences lead models to predict their relationship as "contradiction".

Social stereotypes. In NLP tasks, social stereotypes can be categorized into three primary categories: gender and orientation biases [18, 19], racial and religious biases [18], and political and cultural biases [24]. For instance, some models associate "nurse" more strongly with women [59], or "CEO" predominantly with men [27].

Position bias. Previous studies [29, 40, 41] have shown that LLMs are susceptible to position bias, that is, relying more on information at the beginning, end, or specific positions within the context. For instance, GPT-3.5 and Qwen-2 perform better when the relevant information occurs at the beginning or end, but the performance drops significantly when the information occurs in the middle [25]. In this paper, we focus on the following bias patterns: lexical overlap bias and negation bias, social stereotypes, and position bias.

C Prompt Template Construction

To generate standard prompt templates for bias patterns, we use advanced LLMs (e.g., GPT-4o) to summarize and refine bias patterns in the biased data $\tilde{D}$. The generated prompt templates for bias patterns are tailored to different tasks, as shown in Fig. 8.

To generate counterfactual biased samples, we use the contextual modeling ability of LLMs with task-specific prompts. Fig. 9 presents the prompt templates used for generating these samples.

Table 6: Debiasing performance of *PEDAL* on all subsets of HANS. We report Accuracy.

Backbone		QWen2-7B						Deepseek-V2-7B					
Bias pattern		BASE	LoRA	Δ LoRA	Ext-Sub	<i>PEDAL</i> _{prom}	<i>PEDAL</i> _{data}	BASE	LoRA	Δ LoRA	Ext-Sub	<i>PEDAL</i> _{prom}	<i>PEDAL</i> _{data}
lexical bias	HANS-lex	87.86	86.49	84.78	77.60	96.13	97.98	79.46	78.83	49.65	62.00	91.53	90.97
	HANS-sub	58.28	56.42	53.02	53.75	62.97	60.63	60.95	54.60	50.37	51.25	60.63	62.27
	HANS-con	52.88	58.49	53.10	57.07	74.59	<u>63.48</u>	<u>52.78</u>	52.23	49.57	50.78	<u>53.38</u>	56.20
negation bias	HANS-lex	87.86	86.49	88.00	84.98	<u>92.99</u>	95.56	79.46	78.83	77.11	64.12	<u>85.43</u>	94.70
	HANS-sub	58.28	56.42	60.04	56.12	62.76	60.72	60.95	54.60	55.13	51.06	<u>60.17</u>	60.15
	HANS-con	52.88	58.49	54.85	59.41	68.90	57.95	52.78	52.23	50.29	51.17	<u>54.06</u>	54.76
mixed bias	HANS-lex	87.86	86.49	91.33	89.51	93.61	95.23	79.46	78.83	57.11	61.19	93.94	91.17
	HANS-sub	58.28	56.42	59.53	59.04	60.74	59.14	60.95	54.60	54.98	51.34	62.83	59.43
	HANS-con	52.88	58.49	55.60	57.18	74.07	<u>61.26</u>	<u>52.78</u>	52.23	52.26	50.68	<u>54.54</u>	61.44

Table 7: Overall performance of *PEDAL* based on (IA)³. For HANS, we present the results of the lexical overlap subset.

Bias pattern	lexical overlap bias			negation bias			mixed bias			social stereotypes		position bias	
Method	General MNLI	Unbiased HANS	ANLI	General MNLI	Unbiased HANS	ANLI	General MNLI	Unbiased HANS	ANLI	General BBQ	Unbiased Unqover	General News	Unbiased News
QWen2-7B													
BASE	84.10	87.86	75.39	84.10	87.86	75.39	84.10	87.86	75.39	76.35	65.70	51.35	13.49
CDA	86.01	88.56	84.76	86.12	86.20	84.42	85.61	86.58	84.64	83.23	86.56	/	/
(IA) ³	88.31	78.27	82.17	88.31	78.27	82.17	88.31	78.27	82.17	88.55	83.50	54.36	16.08
Δ (IA) ³	84.17	79.34	85.68	83.62	80.31	80.22	85.42	82.44	84.36	75.12	50.74	50.12	17.68
PEFTDebias	83.95	85.22	74.58	83.95	85.22	74.58	83.95	85.22	74.58	85.47	84.22	51.24	14.53
<i>PEDAL</i> _{prom}	88.19	85.20	86.22	87.55	90.24	84.87	87.69	88.93	85.04	<u>87.20</u>	87.85	<u>54.01</u>	18.53
<i>PEDAL</i> _{data}	86.41	90.99	85.17	85.69	<u>88.93</u>	86.17	86.52	89.44	86.09	86.65	89.05	/	/

Lexical overlap bias

<prompt> Note that there is a lexical overlap bias between the premise and hypothesis. Do not judgetheir relationship based on overlapping words, but rather on their semantics meaning and logic relationship </prompt>

Negation bias

<prompt> Note that there is a negative word bias between the premise and hypothesis, such as 'not', 'no', 'never', 'nothing', etc. Do not judge their relationship solely based on the presence of these negative words, but rather on their semantics meaning and logic relationship. </prompt>

Social stereotypes

<prompt> Analyze the given scenario to identify and evaluate any possible social biases such as gender, race, occupation, age, etc. After conducting the analysis, provide an objective and fair answer to the question posed. Avoid any form of social bias in your final answer. </prompt>

Position bias

<prompt> When summarizing, ensure that you do not rely solely on the beginning information due to positional bias; rather than consider the full breadth of information to accurately capture the expressed meaning. </prompt>

Figure 8: The debiasing prompt template $[P]$ generated for bias patterns in each downstream task.**Lexical overlap bias**

"<prompt> Premise: '...' and Hypothesis: '...' Generating a sentence that contradicts the hypothesis with minimal modifications, and ensuring that words in the generated sentence appear in the premise. Please provide the generated sentence directly. </prompt>"

Negation bias

"<prompt> Premise: '...' and Hypothesis: '...' Generating a sentence that contradicts the hypothesis with minimal modifications, and ensuring that the generated sentence contains negative words such as 'no' or 'not'. Please provide the generated sentence directly. </prompt>"

Social stereotypes

"<prompt> Context: '...' Please read the above text and swap the two main characters along with their associated attributes and descriptions. Ensuring the sentence remains fluent and stylistically consistent with the original. No need to add any additional content, try to ensure consistency with the length of the original text as much as possible. </prompt>"

Figure 9: Prompt templates for generating counterfactual samples in the counterfactual data augmentation strategy.**D Implementation Details of Baselines**

Δ LoRA [56]: Fine-tuning LLMs on tuning data and biased data to obtain LoRA modules: LoRA_{task} and LoRA_{bias} . Then, the debiased LoRA module is derived by computing $\text{LoRA}_{task} - \text{LoRA}_{bias}$, which is merged with the base model to obtain the final debiased model. **Ext-Sub** [13]: Similar to Δ LoRA, the "expert" PEM and "anti-expert" PEM are integrated in the weight direction to achieve debiasing.

PEFTDebias [1]: It consists of two stages: the upstream stage uses PEFT to obtain debiasing parameters after augmenting axis-specific data; the downstream stage incorporates these parameters into the base model and fixes them during full fine-tuning. For feasibility, we omit the full fine-tuning process and directly merge the debiasing parameters with the base model to obtain the final debiased model. **SOD** [25]: SOD is a full fine-tune debiasing method. For feasibility, we modify LoRA's loss function to match SOD's loss function; **ADEPT** [52]: We employ GPT-4o to summarize the neutral and attribute tuples for each social stereotype type in the BBQ dataset. Then, we perform debiasing operations separately for each category using prompt tuning. Finally, we average the debiasing performance across all categories to obtain the overall performance of ADEPT. **FairSteer** [20]: For fairness, we only reproduce the bias reduced effect on the social stereotype provided in the authors' paper.

E Proofs

The output probability distributions of LLM_{FT} and f_{bias} on the sample x_i are denoted as: $P = [p_1, p_2, \dots, p_K]$ and $\bar{P} = [\bar{p}_1, \bar{p}_2, \dots, \bar{p}_K]$, respectively. The adjusted output distribution of LLM_{FT} is $P' = [p'_1, p'_2, \dots, p'_K]$. According to Eq. (7):

$$\sum_{k=1}^K p'_k = \sum_{k=1}^K (p_k + \frac{p_{target} * p_k}{1 - p_{target}}) = \sum_{k=1}^K \frac{p_k}{1 - p_{target}}, \quad (9)$$

Since $p_{target} = 1 - \sum_{k=1, k \neq target}^K p_k$, $\sum_{k=1}^K p'_k = 1$.

Besides, considering the KL divergence:

$$KL(P||\bar{P}) = \sum_{k=1}^K p_k \log \frac{p_k}{\bar{p}_k}, \quad KL(P'||\bar{P}) = \sum_{k=1}^K p'_k \log \frac{p'_k}{\bar{p}_k}, \quad (10)$$

$$\Delta KL = KL(P'||\bar{P}) - KL(P||\bar{P}) = \sum_{k=1}^K (p'_k \log \frac{p'_k}{\bar{p}_k} - p_k \log \frac{p_k}{\bar{p}_k}), \quad (11)$$

According to Eqs. (7) and (11), when $k \neq \text{target}$:

$$\Delta KL = \sum_{k=1}^K \left[\left(p_k + \frac{p_{\text{target}} * p_k}{1 - p_{\text{target}}} \right) \log \frac{p_k + \frac{p_{\text{target}} * p_k}{1 - p_{\text{target}}}}{\bar{p}_k} - p_k \log \frac{p_k}{\bar{p}_k} \right], \quad (12)$$

where target is the biased output. Let $\gamma = \frac{p_{\text{target}} * p_k}{1 - p_{\text{target}}}$:

$$\Delta KL = \sum_{k=1}^K \left[p_k \log \frac{p_k + \gamma}{\bar{p}_k} + \gamma \log \frac{p_k + \gamma}{\bar{p}_k} \right], \quad (13)$$

When $p_k + \gamma \geq \bar{p}_k$, $\Delta KL \geq 0$. For $p_k + \gamma < \bar{p}_k$, since $\log(1+u) \geq \frac{u}{1+u}$:

$$\log \frac{p_k + \gamma}{\bar{p}_k} = \log \left(1 + \frac{\gamma}{\bar{p}_k} \right) \geq \frac{\gamma}{p_k + \gamma}, \quad (14)$$

$$\log \frac{p_k + \gamma}{\bar{p}_k} = \log \left(1 + \frac{p_k + \gamma - \bar{p}_k}{\bar{p}_k} \right) \geq \frac{p_k + \gamma - \bar{p}_k}{p_k + \gamma}, \quad (15)$$

According to Eqs. (13), (14), and (15):

$$\Delta KL \geq \sum_{k=1}^K \left[p_k \frac{\gamma}{p_k + \gamma} + \gamma \frac{p_k + \gamma - \bar{p}_k}{p_k + \gamma} \right] = \sum_{k=1}^K \left[\gamma \left(1 + \frac{p_k - \bar{p}_k}{p_k + \gamma} \right) \right], \quad (16)$$

Since $p_k + \gamma < \bar{p}_k$, $\frac{1}{p_k + \gamma} > \frac{1}{\bar{p}_k}$. According to Eq. (16):

$$\Delta KL \geq \sum_{k=1}^K \gamma \left(1 + \frac{p_k - \bar{p}_k}{\bar{p}_k} \right) = \sum_{k=1}^K \gamma \frac{p_k}{\bar{p}_k}, \quad (17)$$

At this time, $\Delta KL \geq 0$. Thus, adjusting the output probabilities of LLM_{FT} using Eq. (7) drives its output distribution away from that of f_{bias} , thereby achieving the goal of reducing biased outputs.

F More Experimental Results

F.1 Overall performance of PEDAL on the HANS

We provide the overall performance of PEDAL on the unbiased test set HANS, as shown in Table 6. The results show that PEDAL achieves the best debiasing performance across all subsets of HANS.

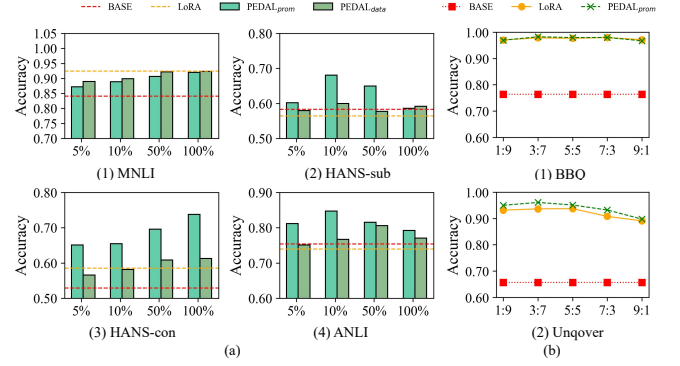


Figure 10: Sensitivity analysis of PEDAL on Qwen2-7B.

F.2 Sensitivity analysis of PEDAL on Qwen2-7B

As shown in Fig. 10(a), PEDAL achieves significant debiasing improvements with only 5% of the tuning data, while obtaining task performance comparable to LoRA. From Fig. 10(b), compared to LoRA, PEDAL exhibits greater robustness to the tuning data composed of larger amounts of biased data, while effectively maintaining task performance of fine-tuned LLMs. The results are consistent with those of PEDAL on Deepseek-V2-7B.

F.3 Performance of PEDAL using IA³

To verify the generality of PEDAL across different PEFT methods, we use another common method, (IA)³, to fine-tune Qwen2-7B, and the results are shown in Table 7. PEDAL outperforms baselines in both preserving task performance and debiasing effectiveness, validating the generality of PEDAL across different PEFT methods.